PULLBACK ALONG A MORPHISM OF RINGED TOPOI PRESERVES THE LOCAL PROPERTIES OF MODULES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Definitions of the local properties of modules on a site. Stacks Project, Modules on Sites, Definition definition-site-local (tag 09B9). Lists the seven local properties whose preservation is proved, and their global counterparts.
- The section on intrinsic properties, reducing ringed-topoi statements to left exact continuous functors. Stacks Project, Modules on Sites, Section section-intrinsic (tag 09BB). Justifies the initial reduction to morphisms of ringed sites with a left exact continuous underlying functor.

Source reference. Original section 03DG: sources/stacks-chow/tags.txt, source lines 4419–4419. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03EF: sources/stacks-chow/tags.txt, source lines 4454–4454. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03DF: sources/stacks-chow/tags.txt, source lines 4418–4418. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03DN: sources/stacks-chow/tags.txt, source lines 4426–4426. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(f, f^\sharp) : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. Let $\mathcal{F}$ be an $\mathcal{O}_\mathcal{D}$-module.*

(1) *If $\mathcal{F}$ is locally free then $f^*\mathcal{F}$ is locally free.*
(2) *If $\mathcal{F}$ is finite locally free then $f^*\mathcal{F}$ is finite locally free.*
(3) *If $\mathcal{F}$ is locally generated by sections then $f^*\mathcal{F}$ is locally generated by sections.*
(4) *If $\mathcal{F}$ is locally generated by r sections then $f^*\mathcal{F}$ is locally generated by r sections.*
(5) *If $\mathcal{F}$ is of finite type then $f^*\mathcal{F}$ is of finite type.*
(6) *If $\mathcal{F}$ is quasi-coherent then $f^*\mathcal{F}$ is quasi-coherent.*

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(7) *If $\mathcal{F}$ is of finite presentation then $f^*\mathcal{F}$ is of finite presentation.*

Proof. According to the discussion in Section 03DG we need only check preservation under pullback for a morphism of ringed sites $(f, f^\sharp) : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ such that f is given by a left exact, continuous functor $u : \mathcal{D} \to \mathcal{C}$ between sites which have all finite limits. Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_\mathcal{D}$-modules which has one of the properties (1) – (6) of Definition 03DL. We know $\mathcal{D}$ has a final object Y and $X = u(Y)$ is a final object for $\mathcal{C}$. By assumption we have a covering $\{Y_i \to Y\}$ such that $\mathcal{G}|_{\mathcal{D}/Y_i}$ has the corresponding global property. Set $X_i = u(Y_i)$ so that $\{X_i \to X\}$ is a covering in $\mathcal{C}$. We get a commutative diagram of morphisms ringed sites

$$\begin{array}{ccc} (\mathcal{C}/X_i, \mathcal{O}_\mathcal{C}|_{X_i}) & \longrightarrow & (\mathcal{C}, \mathcal{O}_\mathcal{C}) \\ \downarrow & & \downarrow \\ (\mathcal{D}/Y_i, \mathcal{O}_\mathcal{D}|_{Y_i}) & \longrightarrow & (\mathcal{D}, \mathcal{O}_\mathcal{D}) \end{array}$$

by Sites, Lemma 03EF. Hence by Lemma 03DF that $f^*\mathcal{G}|_{X_i}$ has the corresponding global property. Hence we conclude that $\mathcal{G}$ has the local property we started out with by Lemma 03DN. □